

Algorithm 4: BIC-cherry + expand via symmetry (weak only)

Require: properly coloured digraph (G, σ) on L with the sicor-in-hub property

Ensure: leaf-coloured network (N, σ)

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1:  $V(N) \leftarrow L \cup \{\rho\}$ ;  $E(N) \leftarrow \emptyset$ 
2: for all  $\{x, y\} \subseteq L$  with  $\sigma(x) \neq \sigma(y)$  do ▷ BIC-cherry network
3:   add  $p_{xy}$  with edges  $(\rho, p_{xy}), (p_{xy}, x), (p_{xy}, y)$ 
4: end for
5: for all  $\{x, y\} \subseteq L$  with  $\sigma(x) \neq \sigma(y)$  do ▷ expansion
6:   for all  $(a, b) \in ((x, y), (y, x))$  with  $(a, b) \notin E(G)$  do
7:      $b' \leftarrow \text{FINDCANDIDATE}(G, a, b)$ 
8:     add new vertex  $q$  with edges  $(p_{xy}, q), (q, a), (q, b')$  ▷  $[ab : ab']$ 
9:     remove edge  $(p_{xy}, a)$  ▷ shortcut
10:  end for
11: end for
12: return  $(N, \sigma)$ 
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13: function FINDCANDIDATE( $G, a, b$ )
14:    $C \leftarrow \{v \in L : \sigma(v) = \sigma(b), (a, v) \in E(G), (v, a) \in E(G)\}$ 
15:   if  $C = \emptyset$  then  $C \leftarrow \{v \in L : \sigma(v) = \sigma(b), (a, v) \in E(G)\}$ 
16:   if  $C = \emptyset$  then  $C \leftarrow \{v \in L \setminus \{b\} : \sigma(v) = \sigma(b), (v, a) \in E(G)\}$ 
17:   if  $C = \emptyset$  then  $C \leftarrow \{v \in L \setminus \{b\} : \sigma(v) = \sigma(b)\}$ 
18:   return random element of  $C$ 
19: end function
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